Clustering and Topic Discovery

POS6933: Computational Social Science

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Overview

- Housekeeping
- Class 6 Problem Set Review
- Discriminating Words
- Clustering
- Topic Models (LDA & STM)

Housekeeping

- Final Paper Drafts Due 3/29 (*Optional*)
- Presentation & Peer Assessment Schedule – Emailed

Class 6 Problem Set Review

- Generally did well
- **Remember**: How you pre-process will have an effect on results downstream!
- **Also...** strength and scope of vocabularies can also impact results!

Discriminating Words

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- **Discrimination Words**: The frequency and distribution of words carry meaningful information about latent categories – i.e., words that appear more frequently in some categories than others are informative signals about that category
- Ex: *Healthcare Provider* vs. *Abortionist* – Terms used in the same debate but relay competing sentiments/connotations/frames

Discriminating Words – Competing Terms Example (D v. R)

Democrats	**Republicans**
Climate	Freedom
Healthcare	Tax
Immigration	Border
Justice	Security
Welfare	Economy

- This is <u>not</u> to say these words invariably represent parties (e.g., useful in vocabulary for dictionary/class assignment)

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- This is <u>not</u> to say these words invariably represent parties (e.g., useful in vocabulary for dictionary/class assignment)
- Rather – Words are useful to help us discriminate (distinguish) between them.
- Ex: Democrats can use *Border*, too (similar or different frames) – but more likely to appear as term used by Republicans.

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- MI $= 0 \sim$ Information between two variables (ex: a label and a word) are completely independent (bad discriminator)
- MI $= 1 \sim$ Knowing about one variable completely informs us of the other (good discriminator).

Calculating Mutual Information

- k = Categories
- j = Word
- π_k = Proportion of Documents for k
- π_{-k} = Probability that Doc Doesn't Belong in k

Calculating Mutual Information (Cont.)

$$H(k) = -\pi_k \log_2 \pi_k - \pi_{-k} \log_2 \pi_{-k},$$

Unconditional Certainty: *How likely we are to assume any random document belongs to a particular category, without even seeing a word*

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$$H(k|j) = -\sum_{k,-k} \sum_{j,-j} \pi_{k,j} \log_2 \pi_{k,j}$$

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$$\mathrm{MI}_{kj} = H(k) - H(k|j)$$

Mutual Information = Unconditional Certainty - Conditional Certainty

Calculating Mutual Information (Example)

Doc ID	Category (k)	Words (j)
1	Red	Apple, Tree
2	Red	Apple, Juice
3	Red	Tree, Green
4	Red	Apple, Green
5	Blue	Blue, Ocean
6	Blue	Ocean, Water
7	Blue	Blue, Water
8	Blue	Ocean, Blue
9	Red	Apple, Tree
10	Blue	Ocean, Tree

Calculating Mutual Information (Example - Cont.)

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$$\pi_{Red} = 5/10 = 0.5$$

$$\pi_{Blue} = 5/10 = 0.5$$

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$$H(K) = -\pi_{Red} \log_2 \pi_{Red} - \pi_{Blue} \log_2 \pi_{Blue}$$

$$H(K) = -0.5 \log_2 0.5 - 0.5 \log_2 0.5$$

$$H(K) = 1$$

Maximum Uncertainty!

Calculating Mutual Information (Example - Cont.)

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Calculate Conditional Certainty For Tree (π_k)

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Calculate Conditional Certainty For Tree (π_k)

$$P(Red|Tree) = 3/4 = 0.75$$

$$P(Blue|Tree) = 1/4 = 0.25$$

$$H(K) = -\pi_{Red}\log_2\pi_{Red} - \pi_{Blue}\log_2\pi_{Blue}$$

Calculating Mutual Information (Example - Cont.)

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$$H(K) = -\pi_{Red}\log_2\pi_{Red} - \pi_{Blue}\log_2\pi_{Blue}$$

$$H(K|Tree) = -(0.75\log_2 0.75 + 0.25\log_2 0.25) = 0.811$$

$$MI(k, Tree) = H(k) - H(k|Tree) = 1 - 0.811 \approx 0.189$$

Calculating Mutual Information (Example - Cont.)

- **What if we tried with *Green*?**

Calculating Mutual Information (Example - Cont.)

- **What if we tried with *Green*?**

  $P(Red|Green) = 2/2 = 1$

  $P(Blue|Green) = 0/2 = 0$

- *Tree* is the only word we can use to discriminate – It's the only term that appears in both categories!

Fightin' Words (Monroe, Coloresi, and Quinn 2008)

- Alternative strategy to identify words that discriminate b/w categories using regularized log-odds ratios: α_j = Dirichlet prior & $W_k^* \sim$ Multinomial(n_k, μ_k)

$$\hat{\mu}_{jk} = \frac{W_{jk}^* + \alpha_j}{n_k + \sum_{j=1}^{j} \alpha_j}$$

- MI identifies words that are most informative about category membership overall – Fightin' Words estimates which words are disproportionately distinctive between groups by accounting for baseline frequencies and sampling variability.
- Estimate corresponds to expected value of posterior distribution $p(\mu|W)$ – $\hat{\mu}_{jk}$ is smoothed probability of word j in category k, allowing for small prior (e.g., Dirichlet).

$$\text{Log Odds Ratio}_{kj} = \log(\frac{\mu_{kj}}{1-\mu_{kj}}) - \log(\frac{\mu - kj}{1-\mu_{-kj}})$$

Fightin' Words (Monroe, Coloresi, and Quinn 2008)

Word	Red	Blue	Total
Apple	4	3	7
Tree	3	1	4
Green	2	0	2
Juice	1	0	1
Blue	0	3	3
Ocean	0	4	4
Water	0	2	2

- Assuming $\alpha_j = 0.5$ for all – Let's do *Apple*

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- Assuming $\alpha_j = 0.5$ for all – Let's do *Apple*
- $\hat{\mu}(Apple|Red) = \frac{n_{(Apple,Red)}+\alpha_{Apple}}{n_{Red}+\sum_j \alpha_j}$
- $\hat{\mu}(Apple|Red) = \frac{4+0.5}{10+7\times 0.5} = 0.33$

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- $\hat{\mu}(Apple|Red) = \frac{4+0.5}{10+7\times 0.5} = 0.33$
- **Let's solve for** $\hat{\mu}(Apple|Blue)$
- $\hat{\mu}(Apple|Blue) = \frac{3+0.05}{13+7\times 0.5} = 0.212$

Fightin' Words (Cont.)

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- $\delta_{\text{Red}} = \log\left(\frac{0.33}{1-0.33}\right) = \log(0.492) \approx -0.71$
- $\delta_{\text{Blue}} = \log\left(\frac{0.212}{1-0.212}\right) = \log(0.269) \approx -1.31$
- $\delta_{kj} = -0.71 - (-1.31) \approx 0.60$

Fightin' Words (Cont.)

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- $\delta_{kj} = -0.71 - (-1.31) \approx 0.60$
- $\exp(0.60) \approx 1.82$ – Meaning that the odds **Apple** appears in **Red** documents is about 1.8 times larger than in **Blue**. The regularization helps ensure this difference is not just due to imbalance in total word counts.

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- Assuming some latent structure (i.e., a discrete grouping or classification) to organize data and explain lexical variance, **clustering algorithms** estimate this structure to represent what is most probable.
- They are **unsupervised** – they infer latent group structure directly from patterns in the data, allowing word usage and co-occurrence to determinate how documents are grouped.
- In essence – they **do not** require pre-defined labels like dictionary or other supervised methods!

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1. Choose groups k – ex: 3 groups of documents, topics, etc.

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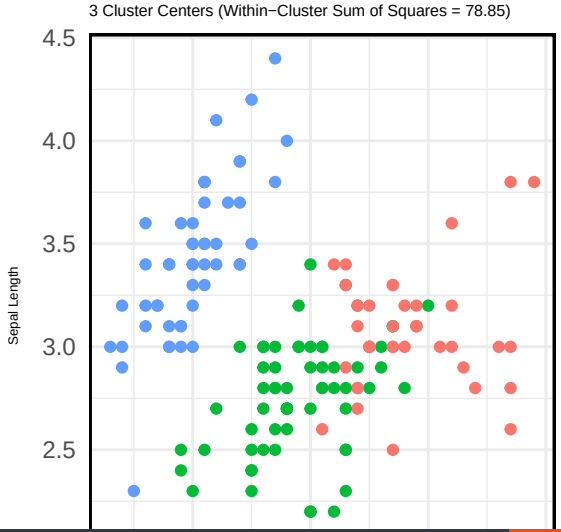
In short... k-means keeps reshuffling items into groups and repositioning the group centers until it finds a stable set of clusters where items in the same group are more similar to each other than to items in other groups.

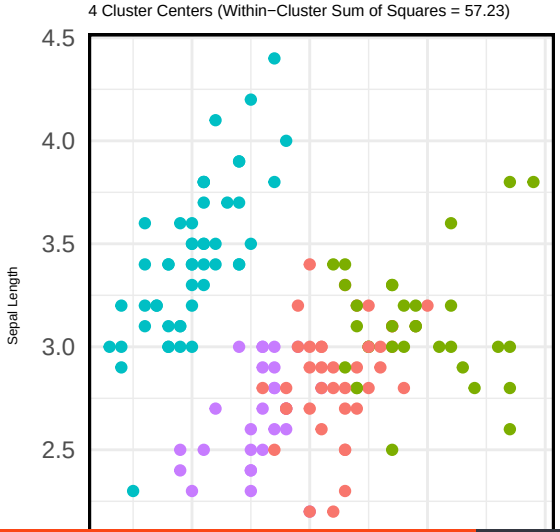
K-Means (Numeric Example)

```r
data <- iris %>%
  rename(sepal_length = Sepal.Length,
         sepal_width = Sepal.Width,
         petal_length = Petal.Length,
         petal_width = Petal.Width) %>%
  select(sepal_length, sepal_width, petal_length, petal_width)

kmeans_example <- kmeans(data, # Iris Data
                         centers = 3, # Cluster Centers (3 Defined)
                         nstart = 25) # No. Independent Runs

kmeans_example$tot.withinss # Within-Cluster Sum of Squares (Dispersion Measure)
```

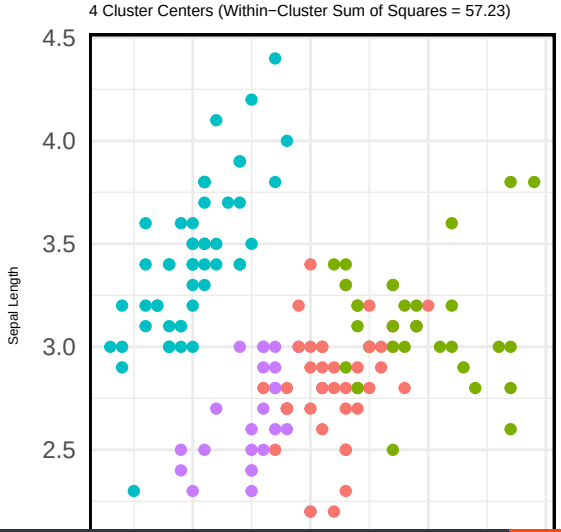
```
[1] 78.85144
```

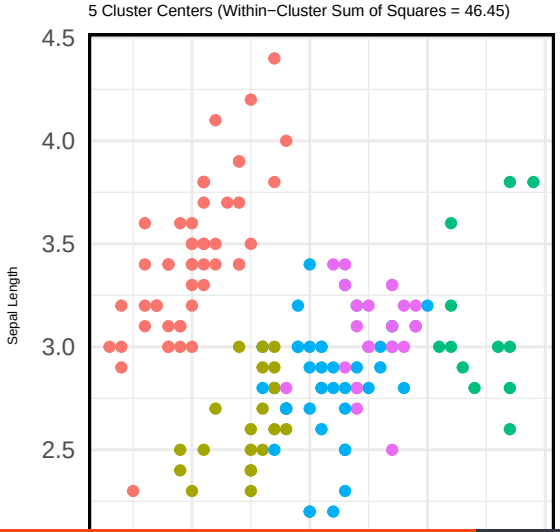
K-Means (Numeric – 3 vs. 4 Cluster Centers)





K-Means (Numeric – 4 vs. 5 Cluster Centers)





K-Means (Text – IMDB Data)

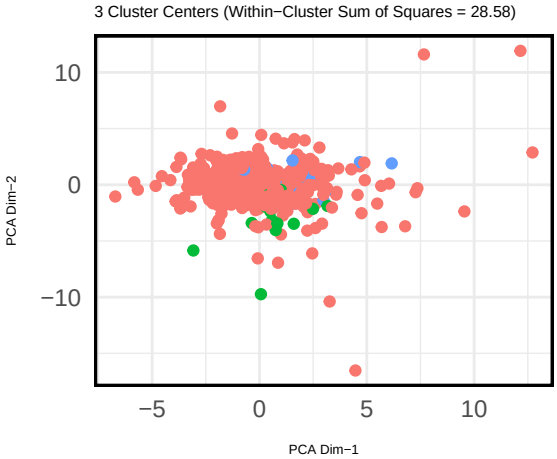
```
imdb_data <- imdb_data$text[1:500] # Only First 500 Rows
imdb_data <- iconv(imdb_data, from = "UTF-8", to = "UTF-8", sub = "") # Fixing Some Characters in IMDB Da
corpus <- VCorpus(VectorSource(sapply(imdb_data, reduce_complexity)))
dtm <- DocumentTermMatrix(corpus, control = list(weighting = weightTfIdf)) # TF-IDF DTM
dtm <- removeSparseTerms(dtm, 0.95)  # (Optional) Reducing Further Sparsity
dtm_matrix <- as.matrix(dtm) # Convert DTM Matrix for K-Means

kmeans_example_3_centers <- kmeans(dtm_matrix, centers = 3, nstart = 25)

kmeans_example_3_centers$tot.withinss
```

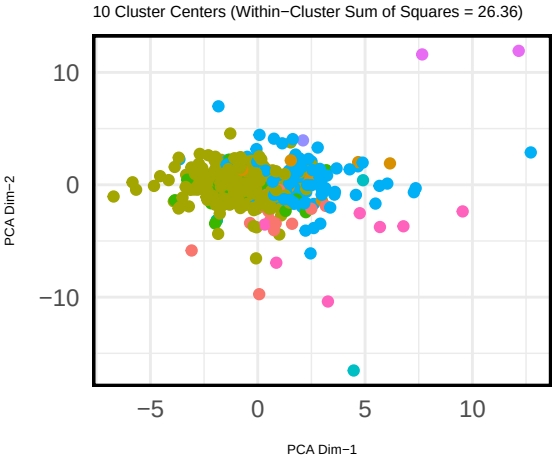
```
[1] 28.58344
```

```
kmeans_example_10_centers <- kmeans(dtm_matrix, centers = 10, nstart = 25)

kmeans_example_10_centers$tot.withinss
```

```
[1] 26.36198
```

K-Means (Text – IMDB Data)





K-Means (Text – IMDB Data) – Notes

- **Remember**: Text data is high-dimensional – not always possible to neatly cluster, especially when models are unsupervised.
- Selection of clusters k motivated by **Theory** and Subsequent **Validation** (multiple mechanisms to measure improvement technically and otherwise – e.g., *does clustering make sense?*)
- Goal is to locate **local optimum** – usually a convergence point where shuffling to minimize sum of squares again yields an improvement less than some determined amount.
- This isn't linear algebra – There can be multiple "correct" solutions...

Clustering – Other Notes

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- **Means v. Mediods**: In the sense of k-means/mediods, its the difference between measuring (a) the location of the average document in the cluster (sensitive to outliers), or (b) the document that minimizes the distance to all others in the cluster (can theoretically omit behavior we're interested in).

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- **Flat v. Hierarchical**: Flat clustering methods (like k-means) produce a single-level partition of the data into a fixed number of clusters. Hierarchical clustering produces a nested tree of clusters, either by successively merging individual observations or recursively splitting the dataset.

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- **Problem**: Clustering algorithms (even with soft partitioning) might obscure results here – speeches (documents) engage with multiple topics, frames, etc – single-member clustering can be problematic.

Topic Models v. Clustering

- **Topic Model**: Rather than assign each document to only one cluster, topic models assign each document with proportional membership to all categories (topics).
- That is...topic models suppose that each document is a mixture across categories – a mixed membership model
- Ex: Re: SOTU Addresses – topic models may help us understand this through the lens of more discrete items like the welfare state, taxation, foreign trade or other topics within the broader scope of discussion related to economic policy.
- We'll focus on two: LDA & STM

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- Note: Like clustering, there is not about estimating the *only* right answer
- Much of the inferential value is assessed post-hoc – recovering useful topics to explain variances in the data recovered through its assignment process.
- Like clustering we can measure (for instance) improvement in log-likelihood to assess whether variance in k are an improvement.

LDA (Latent Dirichlet Allocation)

- Canonical topic model
- Generative probabilistic model – explicitly posits a DGP for documents where documents are mixtures of latent topics and words are drawn conditional on those topics.
- Tries to Recover:
  1. **Document-Topic Distributions**: Each document is represented a distribution over (perhaps multiple) different topics, each with a certain probability.
  2. **Topic-Word Distributions**: Each topic is represented as a distribution over words, meaning that each topic is defined by a set of words (again all with an associated probability of appearing in that topic)

LDA Notation

- Document Index $= i$
- Topic Index $= K$
- Word Position In Document $(i) = m$
- Observed Word $= W_{im}$
- Document-Topic Weights: π_{ik} where $\sum_K \pi_{iK} = 1$ (All Sum to 1)

LDA (Cont.)

$$Z_{im} \sim \text{Multinomial}(1, \pi_i) \quad \text{Topic Indicator for Each Word}$$

$$W_{im} \sim \text{Multinomial}(1, \mu Z_{im}) \quad \text{Observed Word Token For Each Word Position}$$

- Probability of observing a word token in a document ($W_{im} = j$) conditional on both the document-topic weights (π_{iK}) and the topic-word distributions (μ_k) is a weighted average of topic-specific word probabilities, where the weights are given by the document's topic proportions (π_{iK}):

$$p(W_{im} = j \mid \pi_i, \mu) = \sum_K \pi_{iK} \mu_{jK}$$

LDA - Example

```r
library(topicmodels)
data("AssociatedPress")
dtm <- AssociatedPress

lda_model <- LDA(dtm, k = 5, method = "Gibbs", control = list(seed = 1234)) # LDA w/ 5 Topics & Gibbs Sam
topicmodels::terms(lda_model, 10)
```

```
      Topic 1  Topic 2   Topic 3      Topic 4     Topic 5
 [1,] "i"      "percent" "court"      "i"         "government"
 [2,] "people" "million" "police"     "president" "soviet"
 [3,] "two"    "year"    "two"        "bush"      "united"
 [4,] "air"    "billion" "state"      "house"     "states"
 [5,] "years"  "new"     "case"       "new"       "two"
 [6,] "new"    "company" "years"      "committee" "military"
 [7,] "just"   "last"    "federal"    "congress"  "people"
 [8,] "city"   "market"  "department" "dukakis"   "police"
 [9,] "like"   "prices"  "attorney"   "national"  "union"
[10,] "first"  "stock"   "drug"       "campaign"  "officials"
```

LDA - Example (Cont.)

```r
topic_probs <- posterior(lda_model)$topics # Prob Document i Belongs to Topic K
print(round(topic_probs[c(1:10), c(1:5)], 2))
```

```
         1    2    3    4    5
 [1,] 0.20 0.05 0.61 0.06 0.08
 [2,] 0.07 0.31 0.12 0.20 0.30
 [3,] 0.29 0.10 0.44 0.08 0.08
 [4,] 0.30 0.17 0.13 0.14 0.25
 [5,] 0.28 0.14 0.18 0.26 0.15
 [6,] 0.06 0.08 0.08 0.38 0.40
 [7,] 0.22 0.30 0.21 0.11 0.16
 [8,] 0.13 0.13 0.10 0.39 0.25
 [9,] 0.25 0.19 0.18 0.18 0.21
[10,] 0.18 0.13 0.11 0.27 0.30
```

LDA - Example (Cont.)

```r
word_probs <- posterior(lda_model)$terms # Prob Word J in Topic K
head(round(word_probs[, 1:5], 5))
```

```
    aaron abandon abandoned abandoning abbott
1 0.00000 0.00000   0.00039      0e+00  0e+00
2 0.00000 0.00000   0.00000      0e+00  0e+00
3 0.00000 0.00000   0.00000      0e+00  1e-04
4 0.00011 0.00016   0.00001      7e-05  0e+00
5 0.00000 0.00000   0.00006      0e+00  0e+00
```

LDA - Example (Cont.)

- Your turn – do the same with $k = 10$. What changes?

STM - Structural Topic Models

- STMs are another form of topic models that provide for the inclusion of covariates to help explain topic prevalence (X_i^p) and content (X_i^c).

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STM - Structural Topic Models

- STMs are another form of topic models that provide for the inclusion of covariates to help explain topic prevalence (X_i^p) and content (X_i^c).
- For each document – Model first decides how much of each topic is likely to appear (topic prevalence) – this can be driven by features like authorship, publication date, etc.
- For each word in document – Model picks a topic according to those proportions and generates the word from that topic.
- It's a way to combine topic discovery with regression-style analysis, giving you richer insights into both what topics are present and how they're talked about.

STM Example (Setup)

```r
texts <- apply(AssociatedPress, 1, function(x) {
  paste(rep(colnames(AssociatedPress), x), collapse = " ")
})

texts_reduced <- vapply(texts, reduce_complexity, character(1))

corpus_reduced <- VCorpus(VectorSource(texts_reduced))

dtm_reduced <- DocumentTermMatrix( corpus_reduced,
  control = list(wordLengths = c(1, Inf))
)

dfm_ap <- as.dfm(dtm_reduced)

docs_stm <- convert(dfm_ap, to = "stm") # STM Structure

meta <- data.frame(dummy = rep(1, length(docs_stm$documents))) # Document Meta
```

STM Example

```r
stm_model <- stm::stm(
  documents  = docs_stm$documents,
  vocab      = docs_stm$vocab,
  K          = 5,
  prevalence = ~1,
  data       = meta,
  init.type  = "Spectral",
  seed       = 1234,
  max.em.its = 500,
  verbose    = FALSE
)
```

STM Example (Cont.)

```
stm::labelTopics(stm_model, n = 10) # Top Words/Topic
```

```
Topic 1 Top Words:
     Highest Prob: court, year, case, charge, say, two, attorney, go, police, school
     FREX: court, attorney, judge, trial, sentence, justice, lawyer, book, mother, convict
     Lift: male, moore, souter, abrams, abyss, acquit, actress, adjourn, adkins, admission
     Score: court, attorney, sentence, trial, police, lawyer, judge, prison, jury, convict
Topic 2 Top Words:
     Highest Prob: official, report, air, department, state, two, service, test, area, fire
     FREX: water, plane, flight, ship, accident, space, pilot, navy, pentagon, aircraft
     Lift: virus, airplane, aspirin, bay, bomber, cargo, carrier, creek, crow, defect
     Score: plane, flight, pilot, nasa, aircraft, shuttle, airline, accident, ship, pentagon
Topic 3 Top Words:
     Highest Prob: percent, million, year, market, price, company, billion, trade, new, stock
     FREX: market, price, stock, dollar, sale, cent, index, yen, gold, investor
     Lift: chrysler, trader, troy, abboud, acquisition, acustar, aftertax, ag, alkaline, alltime
     Score: stock, cent, percent, price, market, yen, index, rate, dollar, billion
Topic 4 Top Words:
     Highest Prob: soviet, government, official, people, police, state, unite, party, force, military
     FREX: soviet, army, troop, gorbachev, communist, soldier, rebel, israel, africa, israeli
     Lift: abductor, abu, afghanistan, africas, ahmed, alfredo, algeria, algerian, algiers, amal
```

STM Example (Cont.)

```r
topic_word_probs <- exp(stm_model$beta$logbeta[[1]]) #Topic Word Probabilities
head(round(topic_word_probs[,1:5], 5))
```

```
        [,1]    [,2]    [,3]    [,4]    [,5]
[1,] 0.00079 0.00044 0.00032 0.00026 0.00009
[2,] 0.00086 0.00000 0.00000 0.00014 0.00019
[3,] 0.00078 0.00079 0.00029 0.00011 0.00004
[4,] 0.00043 0.00041 0.00012 0.00000 0.00002
[5,] 0.00071 0.00055 0.00021 0.00016 0.00001
```

STM Example (Cont.)

- Your turn – do the same but with different values of topics k.

Next Class

- Distributed Representations of Text (Embeddings)